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Climate variables explain neutral and adaptive variation within salmonid metapopulations: the importance of replication in landscape genetics

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Abstract

Understanding how environmental variation influences population genetic structure is important for conservation management because it can reveal how human stressors influence population connectivity, genetic diversity and persistence. We used riverscape genetics modelling to assess whether climatic and habitat variables were related to neutral and adaptive patterns of genetic differentiation (population-specific and pairwise F_{ST}) within five metapopulations (79 populations, 4583 individuals) of steelhead trout (*Oncorhynchus mykiss*) in the Columbia River Basin, USA. Using 151 putatively neutral and 29 candidate adaptive SNP loci, we found that climate-related variables (winter precipitation, summer maximum temperature, winter highest 5% flow events and summer mean flow) best explained neutral and adaptive patterns of genetic differentiation within metapopulations, suggesting that climatic variation likely influences both demography (neutral variation) and local adaptation (adaptive variation). However, we did not observe consistent relationships between climate variables and F_{ST} across all metapopulations, underscoring the need for replication when extrapolating results from one scale to another (e.g. basin-wide to the metapopulation scale). Sensitivity analysis (leave-one-population-out) revealed consistent relationships between climate variables and F_{ST} *within* three metapopulations; however, these patterns were not consistent in two metapopulations likely due to small sample sizes (N = 10). These results provide correlative evidence that climatic variation has shaped the genetic structure of steelhead populations and highlight the need for replication and sensitivity analyses in land and riverscape genetics.

Keywords: adaptive genetic differentiation, conservation genetics, gene flow, landscape genomic modelling, population genomics, riverscape genetics, salmonids

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Introduction

A main focus of landscape and riverscape genetics is to improve understanding of the effect of environmental variation on patterns of population genetic differentiation (Manel & Holderegger 2013). Climate change is

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predicted to exacerbate existing stressors of habitat fragmentation and degradation on the population genetic connectivity of many terrestrial and aquatics species (Opdam & Wascher 2004; Parmesan 2006; Jaeger *et al.* 2014). The feedback between climate change and fragmented land and river systems might alter evolutionary processes and genetic variation in connected populations by directly influencing gene flow and genetic drift (Dixo *et al.* 2009). Better understanding of the links between spatial patterns of contemporary climate change, habitat variation and genetic diversity is needed to predict future impacts of environmental change on biodiversity (Manel & Holderegger 2013).

Riverscape genetics can reveal how population genetic differentiation is related to and potentially influenced by environmental factors such as stream flow and temperature in freshwater species (Manel *et al.* 2003; Landguth *et al.* 2014). Most often, riverscape genetic studies use putatively neutral patterns of genetic differentiation to describe the interaction between a heterogeneous environment and evolutionary processes such as gene flow, genetic drift and natural selection (Narum *et al.* 2008; Olsen *et al.* 2010). Recent increases in the number of non-neutral (i.e. candidate adaptive) markers have improved the ability to assess patterns of adaptive genetic differentiation in natural populations. Further, inferences about local adaptation based solely on neutral genetic differentiation risk incorrectly identifying the underlying mechanisms driving population differentiation (Landguth & Balkenhol 2012).

Patterns of genetic differentiation are scale-dependent, as the choice of geographic scale (i.e. biological level of organization, such as range-wide or metapopulation) for analysis purposes is important for understanding genetic interactions among populations. This is particularly important in riverscapes in which key habitats vary spatially and temporally across multiple spatial scales (Cushman & Landguth 2010). A common approach in riverscape genetics is to compare patterns of neutral and adaptive genetic differentiation at the metapopulation scale (*sensu* Hanski & Gilpin 1991), because metapopulations are often treated as discrete conservation units for management, monitoring and restoration activities (Rieman & Dunham 2000). Therefore, understanding riverscape controls on gene flow within metapopulations is paramount in understanding local population diversity, productivity and persistence (Hanski & Gilpin 1991; Selkoe *et al.* 2015).

Like choice of scale, the use of replication in riverscape genetics is also an important consideration for improving model robustness. Replication in riverscape genetics (Segelbacher *et al.* 2010; Short Bull *et al.* 2011) is different from replication in the traditional statistical sense and is the comparison of variables selected in riverscape genetic models by correlating genetic differentiation with environmental influences in multiple independent (i.e. nonoverlapping) spatial locations (e.g. metapopulations). Replication is often challenging due to a lack of adequate spatial coverage of sampling, small population sample sizes or limited resources that prohibit conducting genetic analyses in more than a single metapopulation at a time (Moore *et al.* 2011; Short Bull *et al.* 2011). However, replicate observations of similar relationships between environmental variation and genetic differentiation in multiple, spatially discrete metapopulations bolster our mechanistic understanding of environmental influences on local and metapopulation structure. Better understanding these relationships can lead to improved conservation efforts through the increased ability to forecast the impacts of environmental change on genetic diversity (Kovach *et al.* 2015).

Although replication can help identify differences in model selection *among* metapopulations, sensitivity analyses are important for identifying differences *within* metapopulations where results are sometimes dependent on sampling design and available genetic data (Landguth *et al.* 2012). Thus, sensitivity analyses are important for testing the consistency of relationships between environmental variables and genetic differentiation within metapopulations and identifying relationships that are sensitive to the available genetic data (Pannell 1997).

Steelhead trout, the anadromous form of rainbow trout (*Oncorhynchus mykiss*), are an excellent focal species for examining how patterns of environmental variation influence population genetic structure. Steelhead trout occupy a wide geographic range and diversity of freshwater habitats, and their distribution, abundance and phenology is strongly influenced by climatic conditions throughout population-specific adaptations to water temperature and the timing and magnitude of streamflow (e.g. Crozier *et al.* 2008; Jonsson & Jonsson 2009). Thus, ongoing climate change and habitat alternations are expected to differentially affect steelhead populations as streamflow and temperature regimes continue to shift in the coming decades (Mantua *et al.* 2010; Beechie *et al.* 2013).

Recent riverscape genetics studies have primarily focused on broad scale (e.g. analysis across the Columbia River Basin, USA [CRB]) patterns of genetic differentiation and structure in steelhead populations (Blankenship *et al.* 2011; Matala *et al.* 2014). However, few studies have examined patterns of genetic differentiation within and among steelhead metapopulations across heterogeneous river drainages. Here, we used riverscape genetics modelling to assess whether climatic and habitat variables were related to patterns of neutral

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and adaptive genetic differentiation in steelhead populations across the CRB. Our response variables were population-specific and pairwise F_{ST} calculated on sets of 151 putatively neutral loci and 29 candidate adaptive loci describing five steelhead metapopulations. Our independent variables included climate- and habitat-related metrics that we hypothesized would influence genetic differentiation (F_{ST}) within the five metapopulations. We tested the correlative strength of these hypothesized relationships using hierarchal Bayesian modelling and multivariate regression. Finally, we used sensitivity analysis (leave-one-population-out) to further assess the consistency of relationships between climate variables and F_{ST} *within* metapopulations.

Materials and methods

Steelhead metapopulations

We used genetic data from 79 wild populations of steelhead representing five metapopulations within the CRB (Fig. 1). We defined metapopulations on the basis of spatially delineated steelhead conservation units previously determined under the Endangered Species Act (NOAA 2005). Federally recognized distinct population segments (DPSs), the smallest division of a taxonomic species permitted to be conserved under the Endangered Species Act, were delineated based on variables such as reproductive isolation and local adaptation (Waples 1991; Fraser & Bernatchez 2001; Fraser *et al.* 2011). These DPSs were further divided into 16 major population groups (MPGs) in the interior CRB and 2 in the Lower Columbia DPS that delineate metapopulations and areas targeted in conservation planning and research. The 18 MPGs represent regions of similar local populations based on life-history type, geographic location and genetic similarity (ICTRT ICTR 2003; Good *et al.* 2005; Ford *et al.* 2011). Our choice of metapopulations was based on two factors: (i) MPGs are already incorporated in management strategies used by state, federal and tribal agencies, and thus, are relevant to current management practice, and (ii) the genetic baseline used here and in Matala *et al.* (2014) was shown in Matala *et al.* (2014) to 'coincide substantially' with the current delineation of MPGs in the CRB. Currently, 10 DPSs of west coast steelhead in California, Oregon, Washington and Idaho are listed as threatened or endangered under the U.S. Endangered Species Act (NOAA 2005).

We based four of the five metapopulations considered herein on MPGs: Salmon (consisting of 23 populations), Clearwater (22 populations), Eastern Cascades (14 populations) and John Day (10 populations). The Salmon and Clearwater are part of the Snake River Basin DPS in Idaho and Oregon, and the John Day and Eastern Cascades are part of the Middle Columbia DPS in Oregon and Washington. We demarcated the fifth metapopulation, the Lower Columbia, on the basis of an entire DPS because it is similarly sized (spatially) to other MPGs and only consists of 10 local populations (divided into three MPGs).

Genetic data

We used genotypic data spanning years 1996 to 2011 from 4583 adult and juvenile steelhead trout (Table S1, Supporting information). These genetic data were previously used in Matala *et al.* (2014), and full details on genotyping and sequencing can be found there. All individuals were screened for hybridization between steelhead and cutthroat trout (*O. clarkii* spp.), and any hybrids were removed prior to the analysis. Matala *et al.* (2014) describes a set of 191 single nucleotide polymorphisms (SNPs) that were further reduced to 180 after removing loci that were in linkage disequilibrium or deviated from Hardy–Weinberg expectations. Of these 180 loci, Matala *et al.* (2014) identified 24 candidate loci under selection in the four MPGs (Salmon, Clearwater, Eastern Cascades and John Day), and 13 candidate loci in the Lower Columbia DPS populations. We used these previously identified loci in our analysis as our set of candidate adaptive loci. In total, we removed 29 loci from the data set (only 5 of 13 loci in the Lower Columbia set were independent) and used the reduced set (151 loci) as representative of putatively neutral loci. Genetic data files for GESTE and DISTLM *forward* were converted from their original format using PGDSPIDER 2.0.5.2 (Lischer & Excoffier 2012).

Environmental variable data

We hypothesized that neutral genetic differentiation within steelhead metapopulations would be correlated with stream temperature and flow (climate variables) and a suite of physical variables that quantify adult and juvenile steelhead habitat availability (Table 1). Our null hypothesis was that stream distances would be most highly correlated to F_{ST} (i.e. isolation by distance; Wright 1943). Otherwise, correlation between environmental variables and F_{ST} could signal important evolutionary processes (e.g. gene flow and genetic drift) that are affected by the local environment.

Water temperature is considered a key variable influencing steelhead development and patterns in distribution, abundance and genetic diversity (Brannon *et al.* 2004; Richter & Kolmes 2005). Stream temperature data (empirical or modelled) were unavailable for our entire study area, so we used air temperature as a proxy for

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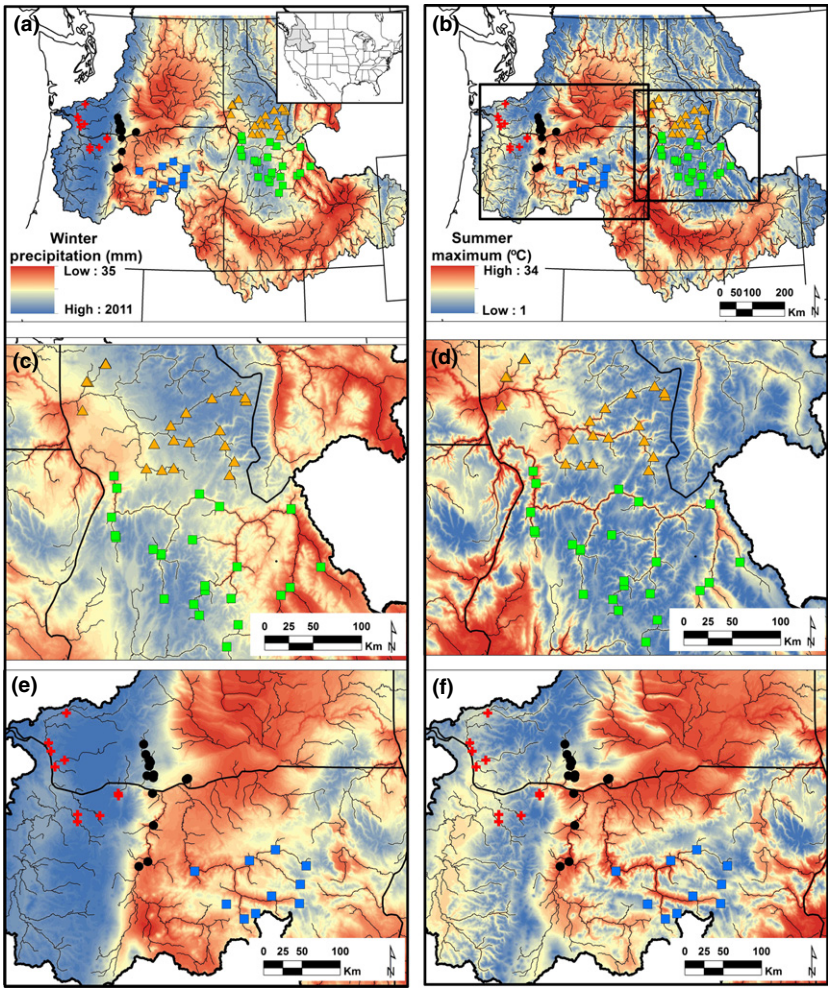

Fig. 1 Map of the five metapopulations of steelhead superimposed on summer maximum temperature and winter precipitation. The top panels depict summer maximum temperature (a) and winter precipitation (b) at the full extent of the Columbia River Basin, USA. Panels (c) and (e) are blowups of the areas delineated in panel (b) with winter precipitation as a background. Panels (d) and (f) are blowups of the same areas with summer maximum temperature as the background. Points represent local populations of steelhead, coloured by their corresponding metapopulation (red crosses = Lower Columbia, black circles = Eastern Cascades, blue squares = John Day, orange triangles = Clearwater, green squares = Salmon).

this metric as air temperatures are correlated with stream temperatures throughout the CRB (Gardner *et al.* 2003; Caissie 2006). Precipitation, one of the drivers of stream flow, has also been shown to be associated with genetic structure of salmonids, including steelhead (Narum *et al.* 2008; Olsen *et al.* 2010; Matala *et al.* 2014). We also used modelled summer and winter flow variables (Wenger *et al.* 2010) in addition to precipitation to assess the influence of flow variation on F_{ST}.

Summer maximum air temperature (15 July–15 September) and winter precipitation (1 January–30 March) data were obtained from gridded (1-km spatial resolution) daily surface meteorological predictions provided by DAYMET (Thornton *et al.* 1997, 2012). Air temperature and precipitation were averaged from 1980 to 1999 to represent the time periods (representing the previous 4–5 generations) corresponding to contemporary patterns in genetic differentiation (F_{ST}). For stream flow variables, we used mean summer flow and winter highest 5% flow events (the number of days between 1 December–28 February with the highest 5% flows for the entire year; Wenger *et al.* 2010). Stream flow, temperature and precipitation variables were averaged for the entire drainage area upstream of each sampling location where spawning and rearing sites were assumed to be (exact sites could not be assigned).

We quantified habitat availability for each metapopulation as follows: (i) total length of critical habitat (as defined by the National Oceanic and Atmospheric Administration under the Endangered Species Act; NOAA 2005); (ii) proportion of critical habitat to total stream length in the drainage area; and (iii) total length of unconfined valley bottoms (Nagel *et al.* 2014). Unconfined valleys are proxies for floodplain habitats that are often preferentially used by salmonids for spawning and juvenile rearing, and their prevalence across space is correlated with patterns of genetic diversity in other salmonid fishes (Kovach *et al.* 2015). Habitat variables were averaged for the entire upstream drainage area. Finally, we included average pairwise stream distances between a single population and all other populations

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Table 1 List of environmental variables used in linear regression against steelhead neutral genetic differentiation (F_{ST})

Environmental variable[†] (description)	Pred.	Biological mechanism(s)	References
Climate-related variables			
Winter precipitation (WINPCP; 1 January–30 March, mm)[‡]	+	Increased streambed scour (rain on snow events) leads to increased embryo deaths (loss of redds) during incubation periods, leading to lower local effective size (N_e) and increased local genetic drift. Differences in hydrology may also select for difference in reproductive timing and therefore genetic differentiation through reproductive isolation in time	Lapointe *et al.* (2000), Hendry & Day (2005), Goode *et al.* (2013) *Data sources:* Thornton *et al.* (1997, 2012), Wenger *et al.* (2010)
Winter highest 5% flow events (WIN5; Number of days from 1 December–28 February in which flows are among the highest 5% for the year)	–	Increased flow events leads to increased straying (dispersal) during adult migration back to spring and summer spawning grounds. Differences in hydrology may also select for difference in reproductive timing and therefore genetic differentiation through reproductive isolation in time	Hendry & Day (2005), Quinn (2005), Olsen *et al.* (2010) *Data sources:* Wenger *et al.* (2010)
Summer maximum temperature (STMAX; 15 July–15 September, °C)[‡]	+	Higher stream temperatures lead directly to increased mortality or indirectly through increased physiological stress during return migration for adults. Thus, increased mortality leads to fewer returning breeders and lower local N_e	MacDonald *et al.* (2000), Quinn (2005), Richter & Kolmes (2005), Farrell *et al.* (2008), Dionne *et al.* (2008), Olsen *et al.* (2010)
	–	Higher stream temperatures lead to blocked routes of migration to natal streams further increasing straying rates leading to lower F_{ST}	*Data sources:* Thornton *et al.* (1997, 2012)
Summer mean flow (SMF; 1 June–30 September, cfs)[‡]	–	Low summer flows impact suitable rearing habitat, increase death by stranding in extremely low flows and can decrease growth rates. Lower summer flows can also act to raise stream temperatures as above, causing returning adults to suffer higher rates of mortality. Thus, overall, increased mortality leads to fewer juvenile or returning breeders, thus lowering local N_e	May & Lee (2004), Harvey *et al.* (2006), Arthaud *et al.* (2010) *Data source:* Wenger *et al.* (2010)
Habitat variables			
Critical habitat length (CH; stream segments designated as critical spawning and rearing habitat, km) Critical habitat proportion (CHP; ratio of critical habitat to total drainage stream length)	–	Local *Ne* increases with the amount of available rearing/spawning habitat and stream carrying capacity, thus lowering local F_{ST}	Bjornn & Reiser (1991), Neville *et al.* (2006), Ozerov *et al.* (2012) *Data sources:* NOAA (2005)
Unconfined Valley bottom (VB; low gradient zones along the river network where floodplains more often occur and support important ecosystem functions, km)	–	Floodplains act as additional thermal refugia and buffer against high flow events. These variables lead to decreased mortality due to streambed scour during incubation, and increased growth rates for juveniles leading to higher local F_{ST}	Stanford *et al.* (2005), Kovach *et al.* (2015) *Data sources:* Nagel *et al.* (2014)
Null hypothesis			
Stream distance (SD; average distance measured along streams between populations, km)	+	Null hypothesis of isolation by distance	Wright (1943), Narum *et al.* (2008)

For each environmental variable, we reviewed the literature on potential biological mechanisms linking the variable to variation in population-specific F_{ST} for neutral loci and made an *a priori* prediction ('Pred.') of the direction of the relationship. References include both the literature reviewed and the data source(s) used to produce each variable used in the analysis. Some variables are proxies for very similar hypothesized relationships such as winter precipitation and winter highest 5% flow events, and all habitat variables (critical habitat length, critical habitat proportion and unconfined valley bottom).
N_e is local effective population size.
[†]Averaged over upstream drainage area.
[‡]Averaged over time.

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within a metapopulation as a variable to test the null hypothesis of isolation by distance (McPhee *et al.* 2007).

GESTE model selection

We used GESTE (GEnetic STructure inference based on genetic and Environmental data; Foll & Gaggiotti 2008), a hierarchical Bayesian model, to determine the best combinations of climatic and habitat characteristics influencing patterns of genetic differentiation within metapopulations. GESTE calculates population-specific values of F_{ST} and uses a general linear model (GLM) to estimate the relationship(s) between environmental variables (climate and habitat) and F_{ST} values. Population-specific F_{ST} is a useful measure of genetic differentiation that accounts for differences in the strength of genetic drift across populations while estimating how divergent each local population is from its designated metapopulation (Balding & Nichols 1995; Foll & Gaggiotti 2006).

All combinations of environmental variables were considered, and model support was evaluated using the reversible jump Markov Chain Monte Carlo (MCMC) algorithm. Model support is indicated in GESTE by the model with the highest posterior probability (Pr) and is the model that best explains the data chosen from all possible combinations of input variables and including the constant or null model. All predictor variables were standardized by subtracting the mean value and dividing by the standard deviation. We used default parameter settings in GESTE, including 10 pilot runs of 5000 MCMC iterations to obtain the parameters of the proposal distributions, followed by a burn in of 50 000 iterations. An MCMC thinning rate of 20 was then used to obtain 10 000 samples from the posterior distributions of each parameter. This approach was replicated three times to confirm model convergence.

Following the methods outlined in Gaggiotti *et al.* (2009), we first conducted an exploratory run using all eight environmental variables in the GESTE model which simultaneously considers all model combinations (Table 1). Subsequently, we limited additional GESTE runs to the top four environmental variables ranked by estimates of overall posterior probability. Overall posterior probabilities were obtained for each variable by summing the posterior probability of all models that included a specific variable in a single GESTE run. Correlation between variables (multicollinearity) can be problematic when the pairwise correlation (i.e. r) between predictor variables exceeds $|0.70|$ (Dormann *et al.* 2013). In the Bayesian framework, this often leads to spurious results due to highly correlated variables, fraction support among several models and incorrect support of the null model, and incorrect parameter estimates (O. Gaggiotti, personal communication). Thus, we used a Pearson's correlation, using the R statistical package (R Core Team 2013), to test for multicollinearity between environmental variables. In those metapopulations where the reduced candidate model set still contained variables that were moderately to highly correlated ($r > 0.50$), we took a more conservative approach and only tested models using a single variable as opposed to the top four variables selected in the exploratory GESTE runs. We followed this same procedure for putatively neutral and candidate adaptive sets of loci. For each run of GESTE, we report highest posterior density interval (HPDI), Pr and the regression coefficient (b).

DISTLM forward model selection

Balkenhol *et al.* (2009) argued there is no optimal approach for riverscape genetics modelling and suggested using at least two complementary modelling approaches. Similar to the approach taken in Olsen *et al.* (2010), we used DISTLM *forward* (distance-based multivariate analysis for a linear model using forward selection; McArdle & Anderson 2001) to conduct a distance-based multivariate multiple regression with forward selection on pairwise matrices of metapopulation genetic distances (F_{ST}) with pairwise matrices of values for environmental variables. We used DISTLM *forward* to first perform marginal tests to estimate the proportion of variance explained independently by each explanatory variable. DISTLM *forward* then conducts conditional tests using a stepwise forward selection procedure that accounts for correlation among variables. Conditional tests select the subset of variables that explain the greatest amount of variation in the response variable (F_{ST}) by fitting the single most informative variable and then subsequently fitting additional variables while holding each previous variable constant. Variables are ranked on the amount of variance explained, and P-values were determined using 9999 random permutations of the genetic distance matrix. We followed this same procedure for putatively neutral and candidate adaptive sets of loci.

Sensitivity analysis

We used sensitivity analyses to identify and highlight uncertainty in the selection of environmental variables correlated with F_{ST} within metapopulations. We iteratively removed data for a single population (i.e. jackknifing) and then recalculated Pearson's correlation (r) values between environmental variables and the population-specific F_{ST} values output by GESTE for the remaining populations within a metapopulation. We followed this same procedure for putatively neutral and

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candidate adaptive sets of loci. We were most interested in changes that resulted in a different choice of environmental variables by our statistical model across different sets of populations when using sensitivity analysis (e.g. leave-one-population-out). High variation of significant variables among replicates would indicate that results are sensitive to the genetic data available from a single population within a metapopulation, and decrease confidence in relationships between environment and F_{ST}.

Results

GESTE model selection

Tests for Pearson's correlation between environmental variables showed some variables were moderately to highly correlated ($r > |0.50|$; Table S2, Supporting information). High Pearson's correlations that led to running single variable models subsequently in GESTE included: (i) summer maximum temperature and winter highest 5% flow events ($r = 0.90$) in the Clearwater; (ii) summer mean flow, winter precipitation and summer maximum temperature ($r \geq |0.59|$) in the Eastern Cascades; (iii) summer maximum temperature and winter precipitation ($r = -0.92$) and summer maximum temperature and summer mean flow ($r = 0.76$) in the John Day; and (iv) critical habitat and unconfined valley bottom with critical habitat proportion ($r = -0.56$ for both) in the Lower Columbia. Thus, for the Clearwater, Eastern Cascades, John Day and Lower Columbia, our GESTE results were based on models considering only a single variable at a time.

Overall, similar climate variables (winter precipitation, summer mean flow, summer maximum temperature and winter highest 5% highest flow events) were selected in multiple metapopulations; however, only in the Eastern Cascades were the same environmental variables correlated with neutral and adaptive genetic differentiation (Table 2). In the Eastern Cascades, winter precipitation and summer mean flow were the only variables supported by both putatively neutral (Pr = 0.61 and 0.60) and candidate adaptive loci (Pr = 0.81 and Pr = 0.78), with similar regression coefficients for both (winter precipitation: $b = 0.69$ and $b = 0.75$; summer mean flow: $b = -0.70$ and $b = -0.79$; Table 2).

For candidate adaptive loci, we found support for summer maximum temperature (Pr = 081; $b = -0.38$; 95% HPDI = −0.65 to −0.1) and winter highest 5% flow events (Pr = 0.51; $b = -0.35$; 95% HPDI = −0.66 to −0.01) in the Clearwater where there was no previous support when testing with putatively neutral loci. In the John Day, there was also support for summer mean flow (Pr = 0.92; $b = -2.04$; 95% HPDI = −4.34 to −0.23) where there was previous support for summer maximum temperature in putatively neutral loci, although these two variables were also significantly correlated ($r = 0.76$). We found no support in the Salmon or the Lower Columbia for a relationship between environmental variables and F_{ST} for candidate adaptive loci despite support of winter precipitation in the Salmon for putatively neutral loci (Pr = 0.68; $b = 0.50$; 95% HPDI = 0.21–0.77).

Table 2 Parameter estimates for the most supported models, using hierarchical Bayesian modelling, associating genetic differentiation (population-specific F_{ST}) with environmental variables for five metapopulations of steelhead from the Columbia River Basin, USA

Metapopulation	Reg. coef.	95% HPDI	Pr	σ^2
Putatively neutral loci				
Salmon				
WINPCP	0.50	0.21; 0.77	0.68	0.44
Clearwater				
	—	—	—	—
Eastern Cascades				
SMF†	−0.70	−1.13; −0.29	1.00	0.60
WINPCP†	0.69	0.27; 1.13	1.00	0.61
John Day				
STMAX†	−0.51	−0.97; −0.09	0.54	0.61
Lower Columbia				
CHP†	−0.43	−0.76; −0.12	0.62	0.35
Candidate adaptive loci				
Salmon				
	—	—	—	—
Clearwater				
STMAX†	−0.38	−0.65; −0.10	0.81	0.26
WIN5†	−0.35	−0.66; −0.01	0.51	0.27
Eastern Cascades				
STMAX†	−0.58	−1.08; −0.10	0.59	0.94
SMF†	−0.79	−1.30; −0.27	0.98	0.78
WINPCP†	0.75	0.22; 1.28	0.95	0.81
John Day				
SMF†	−2.04	−4.34; −0.23	0.92	1.39
Lower Columbia				
	—	—	—	—

Model variables found to be significantly associated are given for each metapopulation along with parameter estimates (the variable regression coefficient and 95% HPDI), model probability (Pr), and the residual variance estimate (σ^2). The 95% HPDI refers to the 95% highest probability density interval (HPDI) around each parameter estimate. A model with complete support has Pr = 1.0, while a model with no support has Pr = 0.0. σ^2 measures the amount of variance not explained by the model. See Table 1 for variable abbreviations and descriptions.
†Denotes GESTE was run with only that single variable.

DISTLM model selection

Putatively neutral and candidate adaptive loci were correlated with the same environmental variables within 3

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of 5 metapopulations (Salmon, Clearwater and Eastern Cascades; Table 3), but not in the others (John Day and Lower Columbia). Winter precipitation most consistently explained variation in pairwise F_{ST} (%var = 17–37%) for putatively neutral and candidate adaptive loci in these metapopulations, despite not being selected in conditional tests (i.e. tests using forward selection). Summer maximum temperature was the second most consistent variable in explaining variation (%var = 22–45%) and significant in marginal tests for putatively neutral and candidate adaptive loci in the Clearwater and the Eastern Cascades, but only selected in conditional tests in the Eastern Cascades. When considering variables that were significant in both marginal and conditional tests, flow-related variables (winter precipitation and winter highest 5% flow events; Table 3) were most often represented in results. Two major exceptions were the correlation of summer maximum temperature by putatively neutral and candidate adaptive loci in the Eastern Cascades and of stream distance, in addition to summer mean flow and winter highest 5% flow events by candidate adaptive loci in the Lower Columbia. The strongest overall signal occurred in the Lower Columbia where variation in pairwise F_{ST} (%var = 99% in combination) in conditional tests was almost entirely explained by summer mean flow, stream distance and winter highest 5% flow events.

We chose to include the results of the marginal tests in DISTLM *forward* for two reasons: (i) we found multicollinearity between variables that confounded results; and (ii) the procedural method use in DISTLM *forward* preferentially chose variables that explained the highest amount of variation which could result in explaining nearly similar amounts of variation, but not the correct underlying mechanisms driving genetic differentiation. For example, DISTLM *forward* identified similar amounts of variance explained in pairwise F_{ST} for summer maximum temperature (%var = 26%), winter precipitation (%var = 24%) and summer mean flow (% var = 22%; Table 3) using marginal tests for the putatively neutral set of loci in the Eastern Cascades. These three variables were also moderately to highly corre-

Table 3 DISTLM *forward* results for each metapopulation and for each set of putatively neutral vs. candidate adaptive loci

Salmon			Clearwater			Eastern Cascades			John Day			Lower Columbia		
Variable	*P*	%var	Variable	*P*	%var	Variable	*P*	%var	Variable	*P*	%var	Variable	*P*	%var
Putatively neutral Loci														
Marginal tests (independent)														
STMAX	0.42	0.04	**STMAX**	**0.00**	**0.31**	**STMAX**	**0.02**	**0.26**	STMAX	0.60	0.06	STMAX	0.84	0.01
SMF	0.06	0.11	**SMF**	**0.00**	**0.22**	**SMF**	**0.01**	**0.22**	SMF	0.98	−0.01	SMF	0.52	0.05
CH	0.66	0.02	CH	0.92	0.01	CH	0.36	0.07	CH	0.90	0.02	CH	0.87	0.01
CHP	0.88	0.01	CHP	0.26	0.06	CHP	0.89	0.03	CHP	0.53	0.07	CHP	1.00	−0.05
VB	0.81	0.02	VB	0.23	0.07	VB	0.42	0.07	VB	0.68	0.04	VB	0.29	0.13
WIN5	0.69	0.02	**WIN5**	**0.00**	**0.39**	WIN5	0.27	0.10	**WIN5**	**0.02**	**0.25**	WIN5	0.36	0.11
WINPCP	**0.00**	**0.20**	**WINPCP**	**0.01**	**0.19**	**WINPCP**	**0.00**	**0.24**	WINPCP	0.28	0.13	WINPCP	0.44	0.09
SD	0.34	0.05	SD	0.07	0.11	SD	0.10	0.15	SD	0.11	0.23	SD	0.61	0.06
Conditional tests (sequential)														
WINPCP	**0.00**	**0.20**	**WIN5**	**0.00**	**0.39**	**STMAX**	**0.02**	**0.26**	**WIN5**	**0.02**	**0.25**	—	—	—
Candidate adaptive Loci														
Marginal tests (independent)														
STMAX	0.13	0.10	**STMAX**	**0.01**	**0.22**	**STMAX**	**0.01**	**0.45**	STMAX	0.36	0.11	STMAX	0.08	0.31
SMF	0.23	0.07	SMF	0.24	0.07	SMF	0.09	0.21	SMF	0.98	−0.08	**SMF**	**0.04**	**0.59**
CH	0.83	0.00	CH	0.45	0.04	CH	0.52	0.04	CH	0.83	−0.01	CH	0.68	0.04
CHP	0.40	0.05	CHP	0.57	0.03	CHP	0.51	0.05	CHP	0.29	0.14	CHP	0.69	0.03
VB	0.89	0.00	VB	0.67	0.02	VB	0.60	0.03	VB	0.82	−0.02	VB	0.95	−0.05
WIN5	0.40	0.05	**WIN5**	**0.00**	**0.35**	WIN5	0.44	0.06	WIN5	0.10	0.31	WIN5	0.24	0.19
WINPCP	**0.01**	**0.22**	**WINPCP**	**0.04**	**0.17**	**WINPCP**	**0.01**	**0.37**	WINPCP	0.14	0.25	WINPCP	0.38	0.12
SD	0.63	0.02	SD	0.14	0.10	SD	0.37	0.08	SD	0.25	0.19	SD	0.13	0.26
Conditional tests (sequential)														
WINPCP	**0.01**	**0.22**	**WIN5**	**0.00**	**0.35**	**STMAX**	**0.01**	**0.45**	—	—	—	**SMF**	**0.04**	**0.59**
												SD	**0.01**	**0.26**
												WIN5	**0.01**	**0.14**

Each test shows the variable, *P*-value (*P*) and the percentage of variation (%var) in genetic differentiation explained by each variable by sequentially using forward selection. Marginal tests are independent tests of each variable, whereas conditional fits each variable sequentially in attempt to explain the maximum amount of variation. All significant results ($P \leq 0.05$) are bolded.

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lated (r > |0.58–0.80|; Table S2, Supporting information), even though DISTLM *forward* only reported summer maximum temperature as important when using conditional tests. However, these three environmental variables were likely not independent and could contribute to neutral and adaptive genetic differentiation; therefore, all three should be considered in reporting results when additional evidence and data are not available.

Sensitivity analyses

Results from the sensitivity analyses showed consistency in model selection (e.g. the same environmental variables selected across leave-one-out population sets) for the Salmon, Clearwater and Eastern Cascades metapopulations. When we iteratively removed a single population at a time from the Salmon metapopulation, we found winter precipitation and summer maximum temperature to be most highly correlated with population-specific F_{ST} for both neutral and adaptive genetic differentiation (Figs 2 and 3). In the Clearwater, the lack of correlation between environmental variables and population-specific F_{ST} remained unchanged for the putatively neutral set of loci, and summer maximum temperature and winter highest 5% flow events consistently remained most highly correlated to population-specific F_{ST} for candidate adaptive loci. In the Eastern Cascades, summer mean flow, winter precipitation and winter highest 5% flow events were similarly correlated to F_{ST} ($r \sim 0.70$) when we removed the Surveyors Creek population for both putatively neutral and candidate adaptive sets of loci.

Sensitivity analyses for the John Day and Lower Columbia revealed uncertainty in top model selection results including shifts in correlation for certain variables. In the John Day metapopulation, removal of the Granite Creek population resulted in critical habitat length and unconfined valley bottom having the strongest association with F_{ST} rather than winter precipitation (with a considerable drop in winter precipitation from $r = 0.63$ to $r = 0.45$) and summer maximum temperature (Fig. 3) for candidate adaptive loci. Of all populations in the John Day, Granite Creek had the largest average winter precipitation and appeared to have a greater impact on the resulting correlation when included in calculations.

For candidate adaptive loci, removing the population from the Upper Mainstem of the John Day resulted in correlation increases ($r = \sim 0.25$ to $r = \sim 0.55$) in these same two variables with a coinciding drop in summer mean flow. In the Lower Columbia, critical habitat and unconfined valley bottom showed similar spikes in correlation values for the putatively neutral set of loci

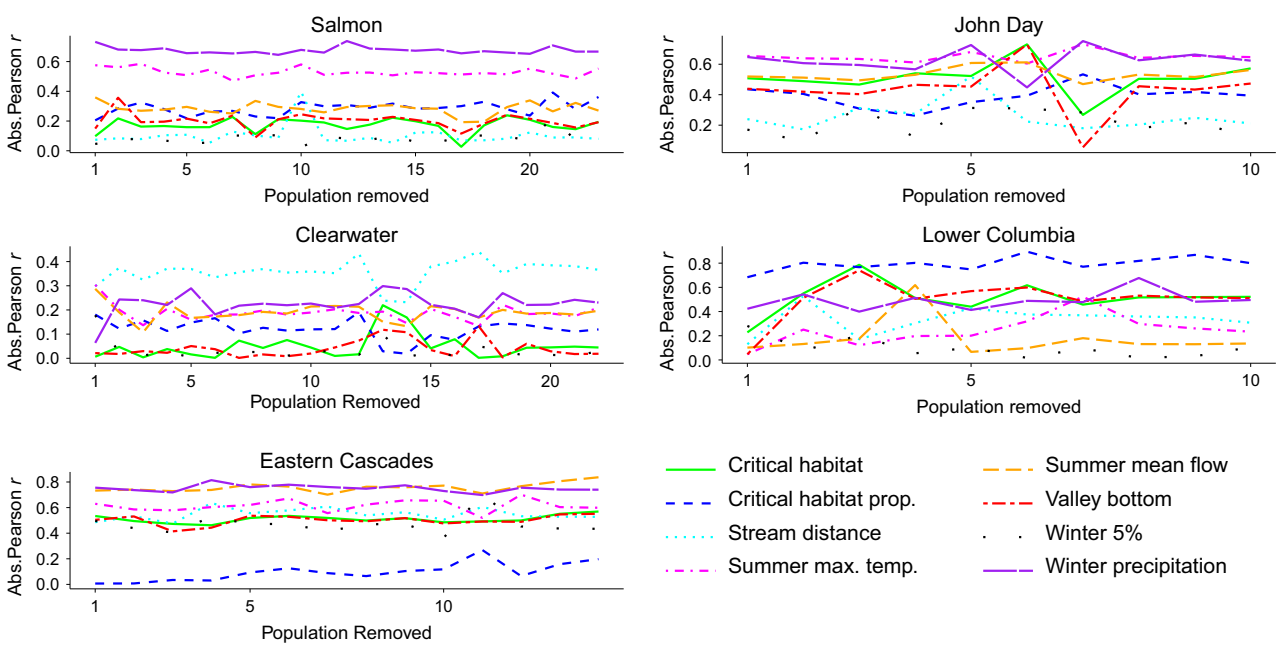

Fig. 2 Results from the sensitivity analyses for each steelhead metapopulations examined using the set of 151 putatively neutral loci. The y-axis is the Pearson r between genetic differentiation (F_{ST}) and each environmental variable. The coloured lines (see legend) depict the change in correlation of each variable when each population is removed (x-axis) during sensitivity analysis. The x-axis is the numbered identifier for the population that was removed during each iteration of the sensitivity analyses.

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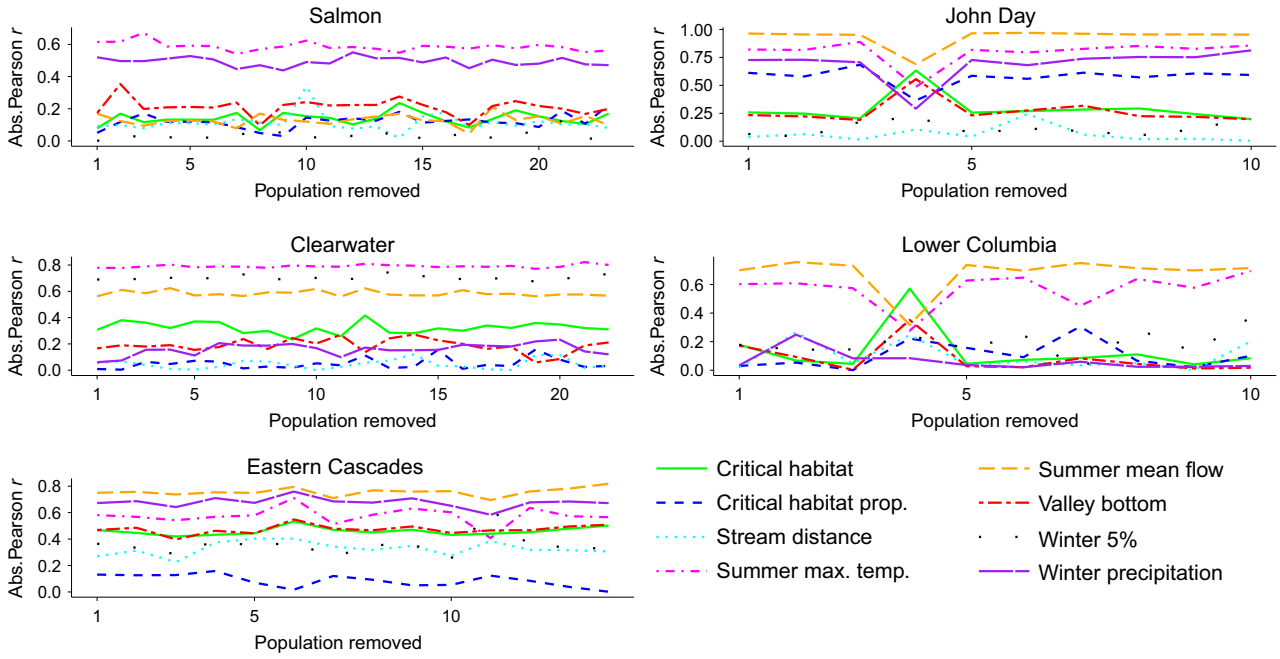

Fig. 3 Results from the sensitivity analyses for all steelhead metapopulations examined using the set of 24 candidate adaptive loci in the Salmon, Clearwater, Eastern Cascades and John Day, and the set of 13 candidate adaptive loci in the Lower Columbia. The *y*-axis is the Pearson *r* between genetic differentiation (F_{ST}) and each environmental variable. The coloured lines (see legend) depict the results for each variable. The *x*-axis is the numbered identifier for the population that was removed during each iteration of the sensitivity analyses.

when removing the East Fork of the Lewis River population where all variables had equal correlation ($r \sim 0.70$). In addition, removing the North Fork of the Lewis River in the Lower Columbia resulted in a shift in the most highly correlated variables with critical habitat replacing summer mean flow; however, this shift was not large enough to cause critical habitat to be selected by either approach (GESTE or DISTLM *forward*).

Discussion

Understanding how environmental variation influences population genetic structure is important for management because it can reveal how human stressors influence persistence of populations of conservation concern. We found that climatic drivers – streamflow and temperature – generally explained neutral and adaptive patterns of genetic differentiation within metapopulations of steelhead trout in the CRB, suggesting that climatic variation likely influences both demographic and genetic processes (e.g. local adaptation). However, we did not observe consistent relationships between climate variables and genetic differentiation across metapopulations, and also, relationships were not always constant within metapopulation when removing a single population from the analysis. Overall, our findings provide correlative evidence that climatic variables influence the genetic structure of steelhead populations and highlight the need for increased use of replication and sensitivity analyses in landscape and riverscape genetic studies.

Replication of land and riverscape genetic studies in multiple, spatially segregated geographic locations and study systems can aid in identifying variability in the influence of environmental variables on genetic structure in different metapopulations and help avoid faulty predictions made by misapplying overgeneralized models across the range of a species (Short Bull *et al.* 2011; Manel & Holderegger 2013). In addition, the use of multiple statistical approaches can also aid in better understanding the generality, consistency and strength of the relationships between environmental variation and genetic differentiation (F_{ST}). Our results indicate that there is considerable spatial heterogeneity in how environmental variation influences evolutionary processes in steelhead trout populations throughout the CRB. Not accounting for the observed heterogeneity in climatic and habitat features explaining genetic differentiation could lead to erroneous predictions of the future effects of climatic change on

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genetic differentiation of certain local populations and metapopulations.

Hypothesized relationships between environmental variables and genetic differentiation

We found that climatic drivers – precipitation and temperature – best explained patterns of genetic differentiation (F_{ST}) of steelhead populations in the CRB, rather than stream distance (isolation by distance) or habitat availability. This general result was consistent when using two complementary statistical approaches (GESTE or DISTLM *forward*) and when considering a set of putatively neutral and candidate adaptive loci.

Winter precipitation was positively correlated with genetic differentiation in the Salmon (for putatively neutral loci) and Eastern Cascades (for putatively neutral and candidate adaptive loci). Winter precipitation was also selected (significant) in the Clearwater, but only in DISTLM *forward* (if considering marginal tests). As hypothesized, snow-dominant hydrology might explain this correlation where peak flows occur during the spring (Hamlet *et al.* 2013) when steelhead embryos are incubating (Table 1). Winter precipitation falling primarily as snow results in increased winter snowpack and peak spring runoff, thereby potentially increasing streambed scour of steelhead redds during embryo incubation periods (late February–June; Goode *et al.* 2013; Matala *et al.* 2014). This could cause high embryo mortality resulting in reduced local effective population sizes (N_e), thereby causing increased genetic drift within populations and *higher* F_{ST} values between populations. Correlation between genetic differentiation and precipitation is consistent with previous studies at large spatial scales for steelhead in the CRB (Narum *et al.* 2008; Blankenship *et al.* 2011; Matala *et al.* 2014).

In the Eastern Cascades, we observed a negative correlation between both summer maximum temperature and summer mean flow and genetic differentiation. High temperatures might cause adult steelhead to stray (i.e. disperse) to streams with colder temperatures when their natal streams are prohibitively warm during migration events (Bradford *et al.* 2009), which has been observed in Chinook salmon (*O. tshawytscha*) populations across the CRB (Westley *et al.* 2015). Streams with decreased flow might further increase genetic differentiation through stranding of migrating steelhead, decreasing rearing habitat and increasing juvenile mortality (May & Lee 2004; Harvey *et al.* 2006; Arthaud *et al.* 2010).

In the Clearwater, summer maximum temperature and extreme winter flooding events were negatively correlated with adaptive genetic differentiation (Table 4). This result could potentially be related to stream temperature and flow regimes during periods of adult migration (August–February; Wade *et al.* 2013). Similar to the hypothesis for the Eastern Cascades, higher temperatures could lead to increased straying of adults and during the later stages of migration, increased flooding events in winter could decrease river stability and increase the dispersal of returning adult steelhead (Quinn 2005). Olsen *et al.* (2010) hypothesized this was the case in Pacific salmon in subarctic Alaskan rivers (Chinook salmon; coho salmon, *O. kisutch;* and chum salmon, *O. keta*). Higher gene flow has also been observed in lake-type (residing in habitat where flooding events are buffered by the lake habitat) vs. river-type sockeye salmon (*O. nerka*; Gustafson & Winans 1999; McGlauflin *et al.* 2011).

Neutral vs. adaptive genetic differentiation

We observed subtle differences between variables correlated with neutral and adaptive genetic differentiation that were mostly test dependent. For example, variables selected in the DISTLM *forward* (when considering marginal tests) tended to be similar or identical across putatively neutral and candidate adaptive sets of loci, but variables selected by GESTE were not (Tables 2 and 3). However, we consistently found temperature and flow-related metrics were often related to genetic differentiation regardless of whether we were using putatively neutral or candidate adaptive loci. Agreement between neutral and

Table 4 Agreement between the GESTE and DISTLM *forward* statistical tests associating environmental variables with genetic differentiation (population-specific F_{ST} and pairwise F_{ST}, respectively) for five metapopulations of steelhead from the Columbia River Basin, USA. Analysis was conducted using a set of putatively neutral (151) and candidate adaptive loci (24)

	Salmon	Clearwater	Eastern Cascades	John Day	Lower Columbia
Putatively neutral Loci	Winter precipitation	None	Winter precipitation; Summer mean flow	None	None
Candidate adaptive Loci	None	Summer maximum temperature Winter highest 5% flow events	Winter precipitation; Summer maximum temperature	None	None

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candidate adaptive patterns of genetic differentiation has also been observed in Atlantic salmon (*Salmo salar*) where neutral markers reflect a similar geographic pattern to markers under selection (Moore *et al.* 2014).

Studies in other salmonids have shown important differences between patterns of neutral and adaptive genetic differentiation. For example, Ackerman *et al.* (2013) found that incorporation of outlier loci to a neutral loci data set changed the group assignment of two populations to nine groups of sockeye salmon identified in Copper River, Alaska. In a comparison of population differentiation detection between neutral and adaptive loci in Chinook salmon, adaptive loci detected significant differentiation between fall and spring run groups, whereas neutral loci detected none (O'Malley *et al.* 2013). Although we uncovered only slight differences in neutral and adaptive patterns of genetic differentiation, we further stress the importance of including putatively neutral and candidate adaptive loci in analyses to increase model reliability and robustness.

The observation that similar environmental variables were often correlated with both neutral and adaptive genetic differentiation might be due to false positives in our candidate adaptive loci (e.g. Lotterhos & Whitlock 2014). However, these results may also suggest that climatic variation can influence both demographic parameters (neutral variation) and natural selection for localized conditions (adaptive variation). For example, major differences in stream flow or thermal regimes often generate strong, genetically based (i.e. locally adaptive) differences in reproductive timing across salmonid populations (e.g. Taylor 1991), which allow populations to exploit advantageous or avoid deleterious environmental conditions. This creates strong reproductive isolation over time, which generates genomic differentiation at putatively neutral and candidate adaptive loci within and among populations (Hendry & Day 2005). In other words, local adaptation in reproductive timing to avoid conditions, such as streambed scouring or high stream temperatures, could produce similar patterns in steelhead genetic differentiation as the impacts of scouring or high temperatures on genetic drift or gene flow (i.e. the hypotheses described above). In reality, these processes likely work hand in hand.

The importance of scale and replication

Our results suggest it is best to avoid extrapolation (or prediction) of results from one metapopulation to another based solely on range-wide studies (e.g. spanning multiple metapopulations or subdrainages). Importantly, salmon population dynamics also exhibit widely varying responses to climatic variation at relatively large and small scales (e.g. Rogers & Schindler 2011; Rogers *et al.* 2013; Kovach *et al.* 2014), a pattern that similarly complicates accurate prediction but ultimately stabilizes long-term population and metapopulation dynamics (Hilborn *et al.* 2003; Schindler *et al.* 2010). Thus, management strategies that identify and protect key environmental processes generating ecological and evolutionary diversity in salmon and steelhead may be more effective than relying on specific predictions of future population or evolutionary dynamics (Schindler & Hilborn 2015).

Similarly, a major challenge when interpreting or applying the results of range-wide analyses is that important idiosyncratic relationships between climatic variation and various aspects of population performance (e.g. recruitment, growth, survival, genetic drift, gene flow, etc.) may be masked (Anderson *et al.* 2010; Rogers & Schindler 2011). Replication is also useful for reducing the effects of migration stressors on modelling results at the metapopulation scale that cannot be ignored at the range-wide scale. For example, steelhead trout within a single metapopulation undertake similar routes of downstream migration and likely occupy similar ocean habitats (Welch *et al.* 2004), making the effects of these variables mostly inconsequential in analysis. Thus, isolating local environmental differences within metapopulations is particularly important. However, it is worth noting that there may be substantial differences between metapopulations related to highly variable migratory distances and unknown ocean distributions for this species (Hayes & Kocik 2014).

Sensitivity analysis

We show how a relatively simple approach to sensitivity analysis can improve identification of variables influencing genetic differentiation while highlighting uncertainties. Results from the Salmon, Clearwater and Eastern Cascades metapopulations qualitatively showed consistent relationships in strength and relative ranking between the most highly correlated environmental variables and population-specific F_{ST} (Fig. 3) when iteratively removing a population. This was likely due in part to the Salmon, Eastern Cascades and Clearwater metapopulations having the most population genetic data.

Sensitivity analysis indicated that results for the John Day and Lower Columbia (both with $N = 10$ populations) were potentially dependent on the relatively limited data used and that removing a single population changed model selection results. Small sample sizes (e.g. the number of populations) are inherently more strongly influenced by single values (or outlier values), and we found that sample sizes in the John Day and Lower Columbia were likely small enough to be problematic when drawing inferences. For example, remov-

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ing the Granite Creek population in the John Day using putatively neutral loci, or the Upper Mainstem using candidate adaptive loci, altered model selection results such that the top model included critical habitat length and unconfined valley bottom (as opposed to winter precipitation). In the Lower Columbia, removing a single population using candidate adaptive loci resulted in critical habitat replacing summer mean flow in relative importance. In both cases, these differences supported a substantially different hypothesis about the mechanism driving genetic differentiation. Thus, identifying these outlier populations is important when identifying environmental drivers influencing genetic variation for an entire metapopulation.

Study limitations and future directions

In some cases, we were limited in our choice or availability of environmental variables. For example, variables likely to have impacted genetic variation in steelhead trout include both historical and contemporary effects of overharvesting, dams, habitat degradation, pollution and hatcheries that were not included in our analysis (Ford *et al.* 2011). Connectivity, flow rates and water temperature in the Columbia River have also been significantly modified since the 1960s by the installation of numerous dams that make up the Federal Columbia River Power System. Construction and operation of these dams have dramatically altered spawning habitat, increased smolt and adult migration mortality, and contributed to significant declines in steelhead and salmon populations (Kareiva *et al.* 2000). In addition, temperatures and flow are impacted by physical habitat characteristics and processes that are difficult to measure such as groundwater exchange that can moderate water temperature at low flows (Stanford *et al.* 2005; Bean *et al.* 2015). Also, our use of critical habitat, although necessary due to a lack of an alternative data source, was an imperfect proxy for actual rearing and spawning habitat for steelhead. Designation of critical habitat does not always include habitat that is currently occupied or used and can include areas where the species is currently absent.

Foll & Gaggiotti (2006) suggest that >10 loci is sufficient power to test for correlations within the GESTE program (e.g. we used 24 and 13 loci in testing for patterns of adaptive genetic differentiation). It is potentially more appropriate in some cases to perform tests using loci individually for identifying correlations to environment variables (e.g. Matala *et al.* 2014). However, we used the same set of tests, requiring the use of more loci, to keep results complementary between putatively neutral and candidate adaptive loci and because we were interested in looking for more widespread patterns across several loci (e.g. across multiple chromosomes; Harrisson *et al.* 2014). There is growing understanding of the role of small-effect loci in adaptation (i.e. polygenic adaptation) where loci behave neutrally but still play a role in adaptation (Harrisson *et al.* 2014). A set of genome-wide neutral loci would be valuable in detecting the subtle effects of polygenic adaptation as salmonids often experience high gene flow, and these small effects have been observed in other salmonid species (Perrier *et al.* 2013; Hecht *et al.* 2015). However, our ability to test for patterns of polygenic adaptation was limited in this study, and future work would benefit from a greater number of genetic markers and testing approaches specific to discovering polygenetic adaptation.

Future work could also explore potential differences in straying rates and effective population sizes of local populations within metapopulations to further illuminate the underlying mechanisms shaping genetic differentiation in each of these metapopulations. Future work might also explore riverscape resistance type models, as we used patch-specific tests that do not test specifically for within metapopulation migration complexity and resistance between populations.

Conclusion

In general, we found a consistent relationship between climate-related variables and genetic differentiation in steelhead trout metapopulations throughout the CRB. Future investigations should consider additional variables (e.g. dams, habitat degradation, pollution, straying of hatchery-reared steelhead into wild populations and life-history variation) that might influence genetic differentiation and connectivity. Our putatively neutral and candidate adaptive loci revealed only small differences in the variables selected in our tests correlating environmental variables with F_{ST}. Replication (using five metapopulations) illuminated important differences between the metapopulation and region-wide scale. Replication in land and riverscape genetic studies is not often conducted, but is critical for better understanding drivers influencing genetic differentiation. Within each metapopulation, sensitivity analyses can help assess the overall robustness of land and riverscape genetic model selection, identify outlier populations and provide understanding of model uncertainty. Our sensitivity analysis suggested two metapopulations (John Day and Lower Columbia) would benefit from additional data (population samples) to draw robust conclusions about the relationship between environmental variables and F_{ST}. Assessing uncertainty using sensitivity analyses and replicate metapopulations will improve general understanding

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and forecasting of genetic variation within and among metapopulations. It will also help advance land and riverscape genetics as a rigorous and predictive discipline for modelling patterns of neutral and adaptive genetic variation, thereby improving conservation of biodiversity.

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References

Ackerman MW, Templin WD, Seeb JE, Seeb LW (2013) Landscape heterogeneity and local adaptation define the spatial genetic structure of Pacific salmon in a pristine environment. *Conservation Genetics*, **14**, 483–498.

Anderson CD, Epperson BK, Fortin M-J *et al.* (2010) Considering spatial and temporal scale in landscape-genetic studies of gene flow. *Molecular Ecology*, **19**, 3565–3575.

Arthaud DL, Greene CM, Guilbault K, Morrow JV (2010) Contrasting life-cycle impacts of stream flow on two Chinook salmon populations. *Hydrobiologia*, **655**, 171–188.

Balding DJ, Nichols RA (1995) A method for quantifying differentiation between populations at multi-allelic loci and its implications for investigating identity and paternity. *Genetica*, **96**, 3–12.

Balkenhol N, Waits LP, Dezzani RJ (2009) Statistical approaches in landscape genetics: an evaluation of methods for linking landscape and genetic data. *Ecography*, **32**, 818–830.

Bean JR, Wilcox AC, Woessner WW, Muhlfeld CC, Tonn W (2015) Multiscale hydrogeomorphic influences on bull trout (*Salvelinus confluentus*) spawning habitat. *Canadian Journal of Fisheries and Aquatic Sciences*, **72**, 514–526.

Beechie T, Imaki H, Greene J *et al.* (2013) Restoring salmon habitat for a changing climate. *River Research and Applications*, **29**, 939–960.

Bjornn T, Reiser D (1991) Habitat requirements of salmonids in streams. *American Fisheries Society Special Publication*, **19**, 83–138.

Blankenship SM, Campbell MR, Hess JE *et al.* (2011) Major lineages and metapopulations in Columbia River *Oncorhynchus mykiss* are structured by dynamic landscape features and environments. *Transactions of the American Fisheries Society*, **140**, 665–684.

Bradford M, Keefer ML, Peery CA, High B (2009) Behavioral thermoregulation and associated mortality trade-offs in migrating adult steelhead (*Oncorhynchus mykiss*): variability among sympatric populations. *Canadian Journal of Fisheries and Aquatic Sciences*, **66**, 1734–1747.

Brannon EL, Powell MS, Quinn TP, Talbot A (2004) Population structure of Columbia River Basin Chinook salmon and steelhead trout. *Reviews in Fisheries Science*, **12**, 99–232.

Caissie D (2006) The thermal regime of rivers: a review. *Freshwater Biology*, **51**, 1389–1406.

Crozier LG, Hendry AP, Lawson PW *et al.* (2008) Potential responses to climate change in organisms with complex life histories: evolution and plasticity in Pacific salmon. *Evolutionary Applications*, **1**, 252–270.

Cushman SA, Landguth EL (2010) Scale dependent inference in landscape genetics. *Landscape Ecology*, **25**, 967–979.

Dionne M, Caron F, Dodson JJ, Bernatchez L (2008) Landscape genetics and hierarchical genetic structure in Atlantic salmon: the interaction of gene flow and local adaptation. *Molecular Ecology*, **17**, 2382–2396.

Dixo M, Metzger JP, Morgante JS, Zamudio KR (2009) Habitat fragmentation reduces genetic diversity and connectivity among toad populations in the Brazilian Atlantic Coastal Forest. *Biological Conservation*, **142**, 1560–1569.

Dormann CF, Elith J, Bacher S *et al.* (2013) Collinearity: a review of methods to deal with it and a simulation study evaluating their performance. *Ecography*, **36**, 27–46.

Farrell AP, Hinch SG, Cooke SJ *et al.* (2008) Pacific salmon in hot water: applying aerobic scope models and biotelemetry to predict the success of spawning migrations. *Physiological and Biochemical Zoology*, **81**, 697–708.

Foll M, Gaggiotti O (2006) Identifying the environmental factors that determine the genetic structure of populations. *Genetics*, **174**, 875–891.

Foll M, Gaggiotti O (2008) A genome-scan method to identify selected loci appropriate for both dominant and codominant markers: a bayesian perspective. *Genetics*, **180**, 977–993.

Ford M, Cooney T, McElhany P *et al.* (2011) *Status review update for Pacific salmon and steelhead listed under the Endangered Species Act: Pacific Northwest*. Draft U.S. Department of Commerce, NOAA Technical Memorandum NMFS-NWFSC-113.

Fraser DJ, Bernatchez L (2001) Adaptive evolutionary conservation: towards a unified concept for defining conservation units. *Molecular Ecology*, **10**, 2741–2752.

Fraser DJ, Weir LK, Bernatchez L, Hansen MM, Taylor EB (2011) Extent and scale of local adaptation in salmonid fishes: review and meta-analysis. *Heredity*, **106**, 404–420.

Gaggiotti OE, Bekkevold D, Jørgensen HBH *et al.* (2009) Disentangling the effects of evolutionary, demographic, and environmental factors influencing genetic structure of natural populations: Atlantic herring as a case study. *Evolution*, **63**, 2939–2951.

Gardner B, Sullivan PJ, Lembo AJ (2003) Predicting stream temperatures: geostatistical model comparison using alternative distance metrics. *Canadian Journal of Fisheries and Aquatic Sciences*, **60**, 344–351.

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Good TP, Waples RS, Adams P (2005) *Updated status of federally listed ESUs of West Coast salmon and steelhead*. U.S. Dept. Commerce, NOAA Tech. Memo. NMFS NWFSC-66.

Goode JR, Buffington JM, Tonina D *et al.* (2013) Potential effects of climate change on streambed scour and risks to salmonid survival in snow-dominated mountain basins. *Hydrological Processes*, **27**, 750–765.

Gustafson RG, Winans GA (1999) Distribution and population genetic structure of river- and sea-type sockeye salmon in western North America. *Ecology of Freshwater Fish*, **8**, 181–193.

Hamlet AF, Elsner MM, Mauger G *et al.* (2013) An overview of the Columbia Basin climate change scenarios project: approach, methods, and summary of key results. *Atmosphere-Ocean*, **51**, 392–415.

Hanski I, Gilpin M (1991) Metapopulation dynamics: brief history and conceptual domain. *Biological Journal of the Linnean Society*, **42**, 3–16.

Harrisson KA, Pavlova A, Telonis-Scott M, Sunnucks P (2014) Using genomics to characterize evolutionary potential for conservation of wild populations. *Evolutionary Applications*, **7**, 1008–1025.

Harvey BC, Nakamoto RJ, White JL (2006) Reduced streamflow lowers dry-season growth of rainbow trout in a small stream. *Transactions of the American Fisheries Society*, **135**, 998–1005.

Hayes SA, Kocik JF (2014) Comparative estuarine and marine migration ecology of Atlantic salmon and steelhead: blue highways and open plains. *Reviews in Fish Biology and Fisheries*, **24**, 757–780.

Hecht BC, Matala AP, Hess JE, Narum SR (2015) Environmental adaptation in Chinook salmon (*Oncorhynchus tshawytscha*) throughout their North American range. *Molecular Ecology*, **24**, 5573–5595.

Hendry AP, Day T (2005) Population structure attributable to reproductive time: isolation by time and adaptation by time. *Molecular Ecology*, **14**, 901–916.

Hilborn R, Quinn TP, Schindler DE, Rogers DE (2003) Biocomplexity and fisheries sustainability. *Proceedings of the National Academy of Sciences of the United States of America*, **100**, 6564–6568.

ICTRT ICTR (2003) *Independent Populations of Chinook, steelhead and sockeye for listed Interior Columbia Basin ESUs*. Interior Columbia Basin Technical Recovery Team Report. July 2003.

Jaeger KL, Olden JD, Pelland NA (2014) Climate change poised to threaten hydrologic connectivity and endemic fishes in dryland streams. *Proceedings of the National Academy of Sciences of the United States of America*, **111**, 1–6.

Jonsson B, Jonsson N (2009) A review of the likely effects of climate change on anadromous Atlantic salmon *Salmo salar* and brown trout *Salmo trutta*, with particular reference to water temperature and flow. *Journal of Fish Biology*, **75**, 2381–2447.

Kareiva P, Marvier M, McClure M (2000) Recovery and management options for spring/summer Chinook salmon in the Columbia River Basin. *Science*, **290**, 977–979.

Kovach RP, Joyce JE, Vulstek SC, Barrientos EM, Tallmon DA (2014) Variable effects of climate and density on the juvenile ecology of two salmonids in an Alaskan lake. *Canadian Journal of Fisheries and Aquatic Sciences*, **71**, 799–807.

Kovach RP, Muhlfeld CC, Wade AA *et al.* (2015) Genetic diversity is related to climatic variation and vulnerability in threatened bull trout. *Global Change Biology*, **21**, 2510–2524.

Landguth EL, Balkenhol N (2012) Relative sensitivity of neutral versus adaptive genetic data for assessing population differentiation. *Conservation Genetics*, **13**, 1421–1426.

Landguth EL, Fedy BC, Oyler-McCance SJ *et al.* (2012) Effects of sample size, number of markers, and allelic richness on the detection of spatial genetic pattern. *Molecular Ecology Resources*, **12**, 276–284.

Landguth EL, Muhlfeld CC, Waples RS *et al.* (2014) Combining demographic and genetic factors to assess population vulnerability in stream species. *Ecological Applications*, **24**, 1505–1524.

Lapointe M, Eaton B, Driscoll S, Latulippe C (2000) Modelling the probability of salmonid egg pocket scour due to floods. *Canadian Journal of Fisheries and Aquatic Sciences*, **57**, 1120–1130.

Lischer HEL, Excoffier L (2012) PGDSpider: an automated data conversion tool for connecting population genetics and genomics programs. *Bioinformatics*, **28**, 298–299.

Lotterhos KE, Whitlock MC (2014) Evaluation of demographic history and neutral parameterization on the performance of FST outlier tests. *Molecular Ecology*, **23**, 2178–2192.

MacDonald JS, Foreman MGG, Farrell AP *et al.* (2000) The influence of extreme water temperatures on migrating Fraser River sockeye salmon (*Oncorhynchus nerka*) during the 1998 spawning season. *Canadian Technical Report of Fisheries and Aquatic Sciences*, **2326**, 117.

Manel S, Holderegger R (2013) Ten years of landscape genetics. *Trends in Ecology & Evolution*, **10**, 614–621.

Manel S, Schwartz MK, Luikart G, Taberlet P (2003) Landscape genetics: combining landscape ecology and population genetics. *Trends in Ecology & Evolution*, **18**, 189–197.

Mantua N, Tohver I, Hamlet A (2010) Climate change impacts on streamflow extremes and summertime stream temperature and their possible consequences for freshwater salmon habitat in Washington State. *Climatic Change*, **102**, 187–223.

Matala AP, Ackerman MW, Campbell MR, Narum SR (2014) Relative contributions of neutral and non-neutral genetic differentiation to inform conservation of steelhead trout across highly variable landscapes. *Evolutionary Applications*, **7**, 682–701.

May CL, Lee DC (2004) The relationships among in-channel sediment storage, pool depth, and summer survival of juvenile salmonids in Oregon Coast Range streams. *North American Journal of Fisheries Management*, **24**, 761–774.

McArdle BH, Anderson MJ (2001) Fitting multivariate models to community data: a comment on distance-based redundancy analysis. *Ecology*, **82**, 290–297.

McGlauflin MT, Schindler DE, Seeb LW *et al.* (2011) Spawning habitat and geography influence population structure and juvenile migration timing of sockeye salmon in the Wood River Lakes, Alaska. *Transactions of the American Fisheries Society*, **140**, 763–782.

McPhee MV, Utter F, Stanford JA *et al.* (2007) Population structure and partial anadromy in *Oncorhynchus mykiss* from Kamchatka: relevance for conservation strategies around the Pacific Rim. *Ecology of Freshwater Fish*, **16**, 539–547.

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Moore JA, Tallmon DA, Nielsen J, Pyare S (2011) Effects of the landscape on boreal toad gene flow: does the pattern-process relationship hold true across distinct landscapes at the northern range margin? *Molecular Ecology*, **20**, 4858–4869.

Moore J-S, Bourret V, Dionne M *et al.* (2014) Conservation genomics of anadromous Atlantic salmon across its North American range: outlier loci identify the same patterns of population structure as neutral loci. *Molecular Ecology*, **23**, 5680–5697.

Nagel DE, Buffington JM, Parkes SL, Wenger S, Goode JR (2014) *A landscape scale valley confinement algorithm: Delineating unconfined valley bottoms for geomorphic, aquatic, and riparian applications*. Fort Collins, CO.

Narum SR, Zendt JS, Graves D, Sharp WR (2008) Influence of landscape on resident and anadromous life history types of *Oncorhynchus mykiss*. *Canadian Journal of Fisheries and Aquatic Sciences*, **65**, 1013–1023.

Neville HM, Dunham JB, Peacock MM (2006) Landscape attributes and life history variability shape genetic structure of trout populations in a stream network. *Landscape Ecology*, **21**, 901–916.

NOAA (2005) *Endangered and Threatened Species; Designation of Critical Habitat for 12 Evolutionarily Significant Units of West Coast Salmon and Steelhead in Washington, Oregon, and Idaho; Final Rule*. U.S. Dept. Commerce, Federal Register. 70 FR 52630. 230.

Olsen JB, Crane PA, Flannery BG *et al.* (2010) Comparative landscape genetic analysis of three Pacific salmon species from subarctic North America. *Conservation Genetics*, **12**, 223–241.

O'Malley KG, Jacobson DP, Kurth R, Dill AJ, Banks MA (2013) Adaptive genetic markers discriminate migratory runs of Chinook salmon (*Oncorhynchus tshawytscha*) amid continued gene flow. *Evolutionary Applications*, **6**, 1184–1194.

Opdam P, Wascher D (2004) Climate change meets habitat fragmentation: linking landscape and biogeographical scale levels in research and conservation. *Biological Conservation*, **117**, 285–297.

Ozerov MY, Veselov AEE, Lumme J, Primmer CRR, Moran P (2012) "Riverscape" genetics: river characteristics influence the genetic structure and diversity of anadromous and freshwater Atlantic salmon (*Salmo salar*) populations in northwest Russia. *Canadian Journal of Fisheries and Aquatic Sciences*, **69**, 1947–1958.

Pannell D (1997) Sensitivity analysis of normative economic models: theoretical framework and practical strategies. *Agricultural Economics*, **16**, 139–152.

Parmesan C (2006) Ecological and evolutionary responses to recent climate change. *Annual Review of Ecology, Evolution, and Systematics*, **37**, 637–669.

Perrier C, Bourret V, Kent MP, Bernatchez L (2013) Parallel and nonparallel genome-wide divergence among replicate population pairs of freshwater and anadromous Atlantic salmon. *Molecular Ecology*, **22**, 5577–5593.

Quinn TP (2005) *The Behavior and Ecology of Pacific Salmon and Trout*. University of Washington Press, Seattle, Washington.

R Core Team (2013) *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Richter A, Kolmes SA (2005) Maximum temperature limits for Chinook, Coho, and Chum salmon, and steelhead trout in the Pacific Northwest. *Reviews in Fisheries Science*, **13**, 23–49.

Rieman B, Dunham J (2000) Metapopulations and salmonids: a synthesis of life history patterns and empirical observations. *Ecology of Freshwater Fish*, **9**, 51–64.

Rogers LA, Schindler DE (2011) Scale and the detection of climatic influences on the productivity of salmon populations. *Global Change Biology*, **17**, 2546–2558.

Rogers LA, Schindler DE, Lisi PJ *et al.* (2013) Centennial-scale fluctuations and regional complexity characterize Pacific salmon population dynamics over the past five centuries. *Proceedings of the National Academy of Sciences of the United States of America*, **110**, 1750–1755.

Schindler DE, Hilborn R (2015) Sustainability. Prediction, precaution, and policy under global change. *Science*, **347**, 953–954.

Schindler DE, Hilborn R, Chasco B *et al.* (2010) Population diversity and the portfolio effect in an exploited species. *Nature*, **465**, 609–612.

Segelbacher G, Cushman SA, Epperson BK *et al.* (2010) Applications of landscape genetics in conservation biology: concepts and challenges. *Conservation Genetics*, **11**, 375–385.

Selkoe KA, Scribner KT, Galindo HM (2015) Waterscape genetics – applications of landscape genetics to Rivers, Lakes, and Seas. In: *Landscape Genetics: Concepts, Methods, Applications* (eds Balkenhol N, Cushman SA, Storfer AT, Waits LP), pp. 220–246. John Wiley & Sons, Ltd, Chichester, UK.

Short Bull RA, Cushman SA, Mace R *et al.* (2011) Why replication is important in landscape genetics: American black bear in the Rocky Mountains. *Molecular Ecology*, **20**, 1092–1107.

Stanford JA, Lorang MS, Hauer FR (2005) The shifting habitat mosaic of river ecosystems. *Travaux*, **29**, 1–14.

Taylor EB (1991) A review of local adaptation in Salmonidac, with particular reference to Pacific and Atlantic salmon. *Aquaculture*, **98**, 185–207.

Thornton PE, Running SW, White MA (1997) Generating surfaces of daily meteorological variables over large regions of complex terrain. *Journal of Hydrology*, **190**, 214–251.

Thornton P, Thornton M, Mayer B *et al.* (2012) Daymet: Daily surface weather on a 1 km grid for North America, 1980–2012. Acquired online (http://daymet.ornl.gov/) in November 2013 from Oak Ridge National Laboratory Distributed Active Archive Center, Oak Ridge, Tennessee, U.S.A.

Wade AA, Beechie TJ, Fleishman E *et al.* (2013) Steelhead vulnerability to climate change in the Pacific Northwest. *Journal of Applied Ecology*, **50**, 1093–1104.

Waples RS (1991) Pacific salmon, *Oncorhynchus* spp., and the definition of "species" under the endangered species act. *Marine Fisheries Review*, **53**, 11–22.

Welch DW, Ward BR, Batten SD (2004) Early ocean survival and marine movements of hatchery and wild steelhead trout (*Oncorhynchus mykiss*) determined by an acoustic array: Queen Charlotte Strait, British Columbia. *Deep Sea Research Part II: Topical Studies in Oceanography*, **51**, 897–909.

Wenger SJ, Luce CH, Hamlet AF, Isaak DJ, Neville HM (2010) Macroscale hydrologic modeling of ecologically relevant flow metrics. *Water Resources Research*, **46**, W09513.

Westley PAH, Dittman AH, Ward EJ, Quinn TP (2015) Signals of climate, conspecific density, and watershed features in patterns of homing and dispersal by Pacific salmon. *Ecology*, **96**, 2823–2833.

Wright S (1943) Isolation by distance. *Genetics*, **28**, 114–138.

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Data accessibility

Genotypes for the set of neutral and adaptive loci, pairwise F_{ST} values for adaptive and neutral loci, the environmental data set and GESTE input files have been archived in http://dx.doi.org/10.5061/dryad.pr065. The genotypes used in this study are a subset of a larger data set archived in http://dx.doi.org/10.5061/dryad.ds6c4 and reported in Matala *et al.* (2014).

Supporting information

Additional supporting information may be found in the online version of this article.

Fig. S1 Box plots showing the distribution of population specific measures of differentiation (F_{ST}) calculated by GESTE for sets of populations within each of five major steelhead metapopulations in the Columbia River Basin, USA.

Table S1 Sample data metadata and variables data used in analysis for each population.

Table S2 Pairwise correlation coefficients (r) reported from the DISTLM forward program for each of the 5 steelhead metapopulations.

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